

Derivatives from Tables

Directions: Use each table to find the indicated derivative. Show all work and simplify your answer.

1. $h(x) = f(x)g(x), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-4	2	2	1
2	-1	2	1	-5
3	-4	6	-1	4
6	8	1	-4	-4

2. $h(x) = \frac{f(x)}{g(x)}, h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	6	-4	2	3
2	8	2	-2	1
3	8	-1	3	-2
6	6	4	-2	-5

3. $h(x) = \frac{2f(x)}{3g(x)}, h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	7	4	4	5
2	-6	2	-1	-1
3	-3	-3	-2	-1
6	-2	5	-2	-4

4. $h(x) = f(g(x)), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-7	6	-4	3
2	-8	2	-1	2
3	7	5	-4	5
6	7	3	2	-1

5. $h(x) = f(3x), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-6	-2	-5	-3
2	8	3	3	-1
3	7	-5	5	-1
6	8	-6	-1	-3

6. $h(x) = f(g(2x)), h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	6	2	-1	2
2	-6	3	-4	-2
3	1	-4	3	-1
6	4	-4	-5	2

7. $h(x) = g(f(2x)), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-7	2	-1	-5
2	3	2	-4	1
3	-6	-5	-5	4
4	3	4	-1	-2
6	3	-1	2	-4

8. $h(x) = f(x^2), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	6	2	5	-5
2	-5	3	4	-5
3	2	5	-1	5
4	-2	2	-2	-2
6	8	5	5	-4

9. $h(x) = \frac{f(5x)}{g(x)}, h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-5	-6	-2	-1
2	-7	3	1	1
3	4	-3	-1	-5
6	2	-3	3	-4
10	-6	4	-2	2

10. $h(x) = f(3x)g(x), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	-1	1	1
2	8	3	-3	1
3	2	-1	4	5
6	2	-2	-2	4

11. $h(x) = f(5x) - x^2, h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-1	3	3	-5
2	7	4	1	2
3	7	-4	2	-3
5	-4	-1	-2	-4
6	1	3	5	5

12. $h(x) = g(f(2x)) - 2x, h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	5	2	4	-1
2	4	3	-3	-3
3	6	2	-4	-5
4	1	2	-3	-4
6	8	-4	-4	-5

13. $h(x) = \frac{f(2x)}{g(x)} - 5x, h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	7	-5	-4	-3
2	-5	3	5	-2
3	-6	-4	-4	-2
4	2	-4	4	1
6	-2	-1	-2	-3

14. $h(x) = f(2x)g(4x), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	3	-3	-1
2	-6	3	-2	-3
3	-7	-3	4	3
4	-3	6	2	2
6	8	3	-5	-1
8	4	-3	4	3

15. $h(x) = \frac{f(2x)}{g(4x)}, h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	1	3	-1	5
2	-5	-6	-4	4
3	-3	2	1	5
4	7	2	4	-5
6	-8	-6	-4	-1

16. $h(x) = f(g(x^2)), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	-6	-1	3
2	4	3	3	-3
3	-4	1	-5	5
4	3	3	-4	-2
6	-1	-5	2	-5

17. $h(x) = f(g(2x)), h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-1	2	-1	-2
2	-5	2	-3	1
3	4	-6	3	-1
6	-2	-4	4	-5

18. $h(x) = g(f(2x)) - 2x, h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	7	1	-2	3
2	1	3	5	-4
3	4	6	-2	-1
4	3	-5	-4	-3
6	4	5	3	-1

19. $h(x) = f(3x)g(4x), h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-3	2	-5	3
2	-3	-2	3	-2
3	3	-3	1	-5
4	1	-5	2	5
6	-7	-1	-3	3

20. $h(x) = \frac{f(4x)}{g(x)} - 3x^2, h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-4	-2	1	5
2	-1	1	3	-1
3	-2	1	4	5
4	6	6	1	3
6	6	6	-3	1